

Motivation

Problem: consider data $x_{1:M} = (x_1, \dots, x_M) \in \mathcal{X}$ generated from unknown distribution P_0 and the class of models $\mathcal{M}_\Theta = \{P_\theta : \theta \in \Theta\}$. Priors $\Pi_\eta \in \mathcal{P}(\Theta)$.

Goal: learn the best possible predictive $P_q = \int P_\theta q(d\theta)$ using scoring rules and best possible η for models which can be well-specified $P_0 \in \mathcal{M}_\Theta$ and miss-specified $P_0 \notin \mathcal{M}_\Theta$.

Free energy

Approach: find the best possible (q, η) by solving the problem

$$\mathcal{F}_M^{PrO}(\eta, q) = \int \log \left(\frac{q(\theta)}{\pi_\eta(\theta)} \right) q(d\theta) + \frac{\lambda_M}{M} \sum_{i=1}^M S \left(\int p(x_i|\theta) q(d\theta) \right) \quad (0.1)$$

where S is a scoring rule (e.g. $S = -\log dP$). We call these Predictively Oriented Posteriors (PrO).

Compare with the usual approach in Bayesian statistics where one solves

$$\mathcal{F}_M^{GB}(\eta, q) = \int \log \left(\frac{q(\theta)}{\pi_\eta(\theta)} \right) q(d\theta) + \frac{\lambda_M}{M} \sum_{i=1}^M \int S(p(x_i|\theta)) q(d\theta) \quad (0.2)$$

of which Gibbs posteriors are minimisers of.

Using Jensen's inequality one sees $\mathcal{F}_M^{GB}(\eta, q) \geq \mathcal{F}_M^{PrO}(\eta, q)$.

Solution

Optimisation: To solve (0.1) use joint gradient descent on (q, η) using metric considered for the space $\mathcal{M}_2 = \mathbb{R}^{d_\eta} \times \mathcal{P}_2^1(\mathbb{R}^{d_\theta})$ is

$$d((\eta, q), (\eta', q')) = \sqrt{d_E(\eta, \eta')^2 + d_{W_2}(q, q')^2} \quad (0.3)$$

Optimise (0.2) and (0.1) on $\mathcal{M}_2$ via (joint) gradient flow

$$\text{grad}_{\mathcal{M}_2} \mathcal{F}_M^{\text{PrO}}(\eta, q) = (\text{grad}_{\mathbb{R}^{d_\eta}} F(\eta, q), \text{grad}_{\mathcal{P}_2(\mathbb{R}^{d_\theta})} F(\eta, q))$$

Mean Field interacting particle system

$$\Theta_{k+1}^n = \Theta_k^n + h \left[\nabla_\theta \log p_{\eta_k}(\Theta_k^n) + \frac{\lambda_M}{M} \sum_{i=1}^M \frac{\delta}{\delta q} S \left(\frac{1}{N} \sum_{m=1}^N p(x_i|\Theta_k^m) \right) \right] + \sqrt{2h} W_n \quad (0.4)$$

$$\eta_{k+1} = \eta_k + \frac{h}{N} \sum_{n=1}^N \nabla_\eta \log p_{\eta_k}(\Theta_k^n) \quad (0.5)$$

First order optimality condition

The joint gradient $\text{grad}_{\mathcal{M}_2} F(\theta, q) = 0$ if and only if

$$q_{M,\eta}^*(\theta) \propto \frac{\lambda_M}{M} \pi_\eta(\theta) \exp \left(\sum_{i=1}^M \frac{\delta S(P_q(\cdot), y_i)}{\delta q}(\theta) \right) \Big|_{q=q_{M,\eta}^*}$$

and

$$\frac{\partial}{\partial \eta} F_M^{\text{PrO}}(\eta) = \frac{\partial}{\partial \eta} \mathcal{F}_M^{\text{PrO}}(q, \eta) \Big|_{q=q_{M,\eta}^*} = - \int \nabla_\eta \log(\pi_\eta(\theta)) q_{M,\eta}^*(\theta) d\theta = 0$$

Some results

1. **Existence and Uniqueness**, for fixed η the $q_{M,\eta}^*$ exists and is unique under appropriate assumptions.
2. **Interaction term**, the PrO has an interacting term rather than just a drift.
3. **Convergence of minimisers**, the q minimisers of (0.1) $q_{M,\eta}^* \rightarrow q^*$ for fixed η .

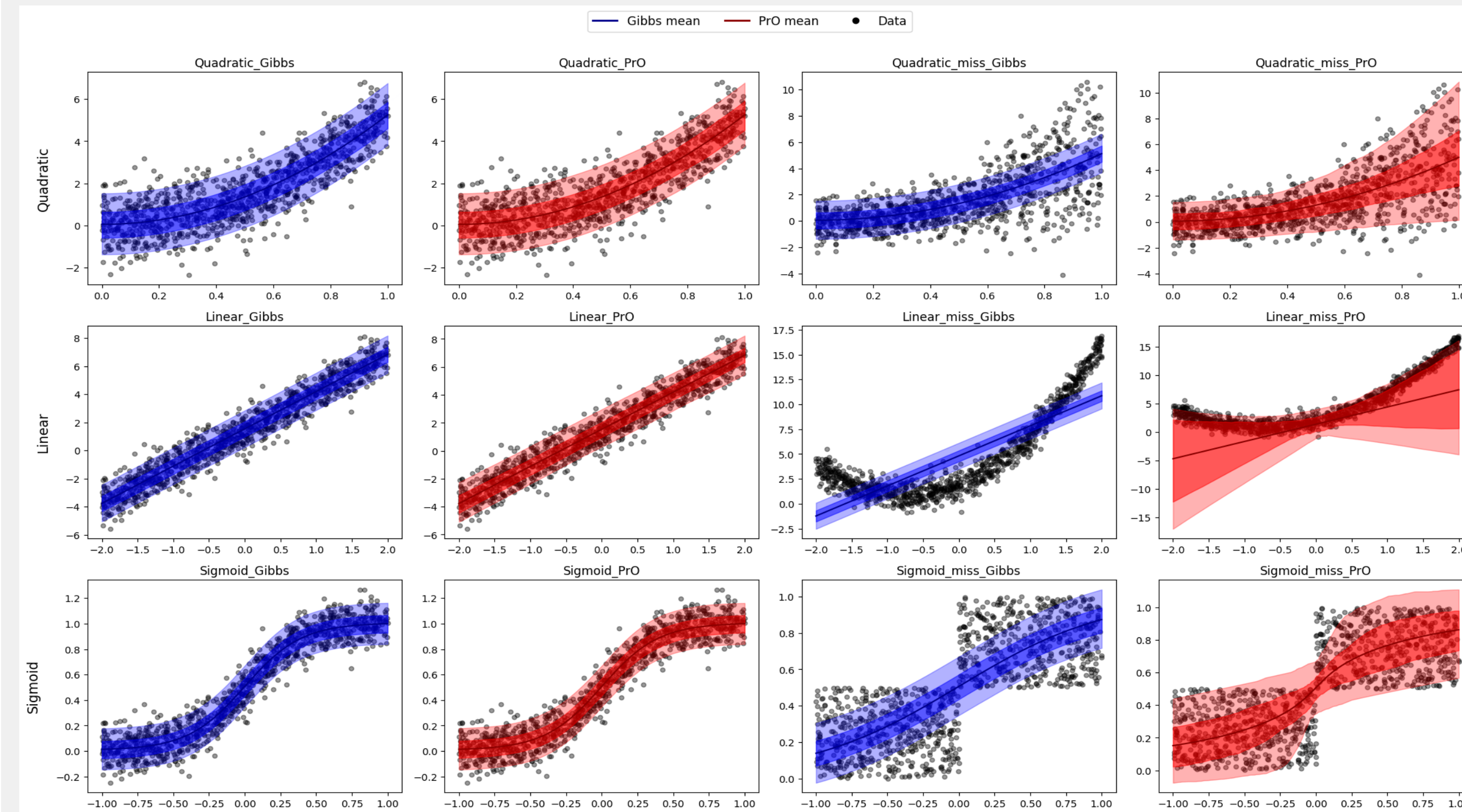
Hierarchical toy examples

Some experiments

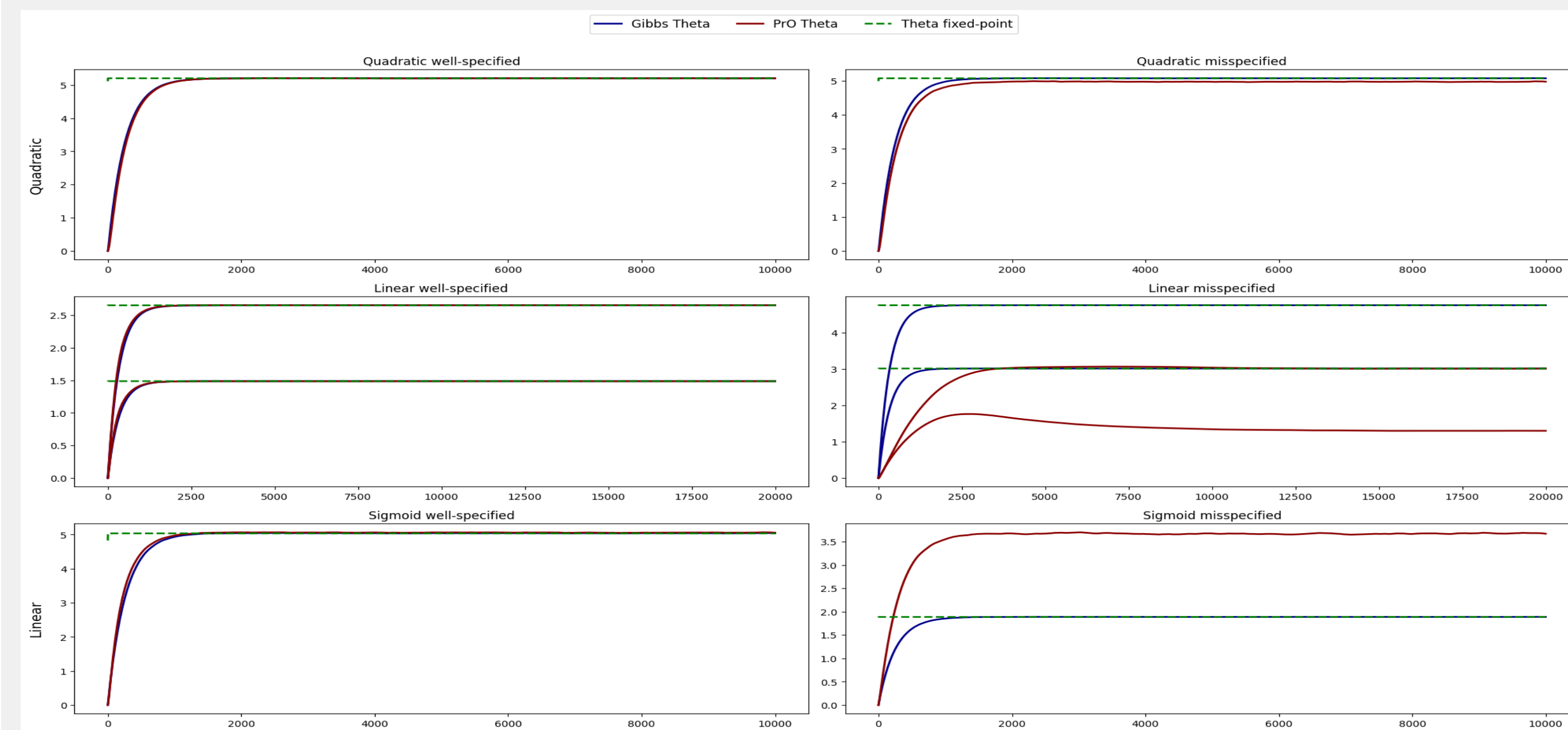
$$x_i = f_\theta(z_i) + \varepsilon_{x,i}, \quad \theta = \eta + \varepsilon_\theta \quad (0.6)$$

with the PGD Algorithm in the well specified cases and miss-specified cases.

1. Quadratic, $f_\theta(z) = \theta^2 z$
2. Linear, $f_\theta(z) = \theta_1 + z\theta_2$
3. Sigmoid, $f_\theta(z) = \frac{1}{1+\exp(-\theta z)}$



Experiments well specified (left) and miss-specified (right) toy models with different f predictives

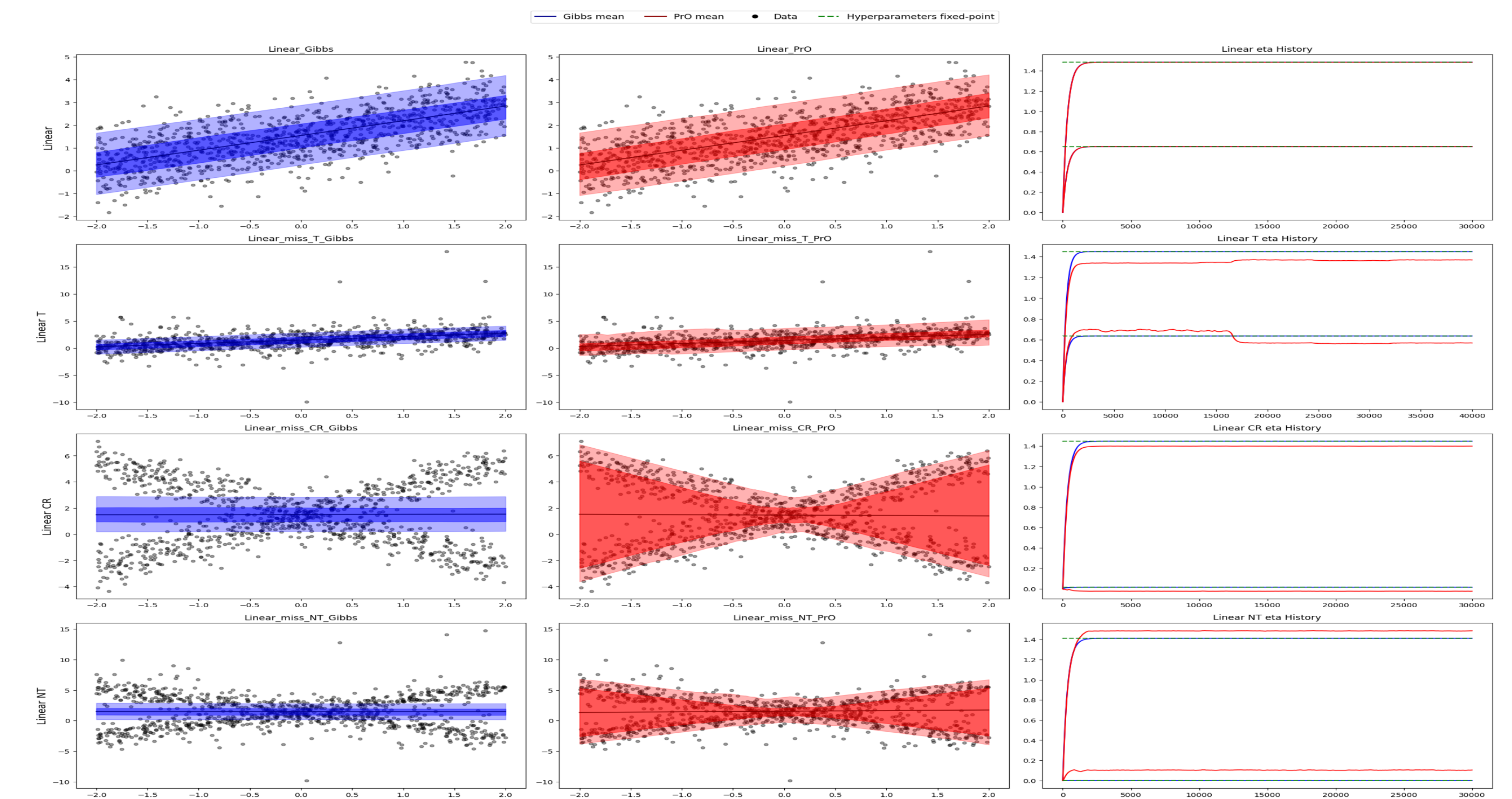


Experiments well specified (left) and miss-specified (right) hyperparameter

Different miss-specified data

For the hierarchical model linear regression (0.6).

1. well specified data i.e. $\varepsilon_{x,i} \stackrel{iid}{\sim} \mathcal{N}(0, \sigma_x^2)$.
2. heavy-tailed noise $\varepsilon_{x,i} \stackrel{iid}{\sim} t_{df}(0, \sigma^2)$ where df is degrees of freedom
3. $\theta_2 = 2 \cdot (-1)^{\xi_i}$, with $\xi_i \stackrel{iid}{\sim} \text{Ber}\left(\frac{1}{2}\right)$
4. both $\varepsilon_{x,i} \stackrel{iid}{\sim} t_{df}(0, \sigma^2)$ and $\theta_2 = 2 \cdot (-1)^{\xi_i}$, with $\xi_i \stackrel{iid}{\sim} \text{Ber}\left(\frac{1}{2}\right)$



Experiments well specified (left) and miss-specified (right) toy models with different miss-specification

Bounds

Well-specified: $\theta_0 = \text{argmin}_{\theta \in \Theta} \text{KL}(P_0 \| P_\theta)$

$$\mathbb{E} \left[\text{KL} \left(P_{\theta_0} \| P_{q_{M,\hat{\eta}_M}^*} \right) \right] \leq \inf_{\eta} \left\{ r_M + \frac{d_S(\eta)}{M} \log \left(\frac{e M C_S(\eta)}{d_S(\eta)} \right) \right\}$$

Trivial miss-specified: $\theta^* = \text{argmin}_{\theta \in \Theta} \text{KL}(P_0 \| P_\theta)$

$$\mathbb{E} \left[\text{KL} \left(P_{\theta_0} \| P_{q_{M,\hat{\eta}_M}^*} \right) \right] \leq \inf_{\eta} \left\{ \text{KL}(P_0 \| P_{\theta^*}) + r_M + \frac{d_S(\eta)}{M} \log \left(\frac{e M C_S(\eta)}{d_S(\eta)} \right) \right\}$$

Non trivial miss-specified: $q^* \in \text{arginf}_{q \in \mathcal{P}(\Theta)} \text{KL}(P_0 \| P_q)$ such that $\inf_{\eta} \text{KL}(q^* \| \pi_\eta) < \infty$

$$\mathbb{E} \left[\text{KL}(P_0 \| P_{q_{M,\hat{\eta}}^*}) \right] < \mathbb{E} \left[\text{KL}(P_0 \| P_{\dagger_{M,\hat{\eta}}}) \right] \quad (0.7)$$

$$\mathbb{E} \left[\text{KL} \left(P_{\theta_0} \| P_{q_{M,\hat{\eta}}^*} \right) \right] - \text{KL} \left(P_0 \| P_{q^*} \right) \leq \inf_{\eta} \left\{ \frac{1}{M} \text{KL}(q^* \| \pi_\eta) + r_M \right\} \quad (0.8)$$

References

- [1] Juan Kuntz, Jen Ning Lim, and Adam M. Johansen. Particle algorithms for maximum likelihood training of latent variable models. In *Proceedings of The 26th International Conference on Artificial Intelligence and Statistics*, volume 206 of *Proceedings of Machine Learning Research*, pages 5134–5180, 2023. URL <https://proceedings.mlr.press/v206/kuntz23a.html>.
- [2] Qingyang Liu, Matthew A. Fisher, Zheyang Shen, Katy Tant, Xuebin Zhao, Andrew Curtis, and Chris. J. Oates. Detecting model misspecification in bayesian inverse problems via variational gradient descent, 2025. URL <https://arxiv.org/abs/2512.01667>.
- [3] Yann McLatchie, Badr-Eddine Cherief-Abdellatif, David T. Frazier, and Jeremias Knoblauch. Predictively oriented posteriors, 2026. URL <https://arxiv.org/abs/2510.01915>.